File-System Consistency

1 Basic Idea

Definition

File-system consistency means that disk blocks, i-nodes, directories, free-space data, and link counts agree with each other.

Why It Breaks

File systems often modify blocks in memory and write them later. A crash before all writes finish can leave metadata inconsistent.

2 Critical Metadata

Metadata	Why it matters
i-node blocks	Store file ownership, link count, and disk block addresses.
Directory blocks	Map file names to i-node numbers.
Free list or bitmap	Records which disk blocks are free.
Data block pointers	Decide which file owns each disk block.

3 Consistency Checkers

Checker

A consistency checker, such as UNIX `fsck`, scans a file system after a crash and repairs metadata contradictions.

- Each partition is checked independently.
- The checker uses file-system redundancy to detect errors.
- Two main checks are block consistency and file/link consistency.

4 Block Consistency

Method

Build two counters for every disk block: one for appearances in files and one for appearances in the free list or free bitmap.

1. Read all i-nodes using raw disk access.
2. For every block used by a file, increment its "in use" counter.

3. Scan the free list or bitmap.
4. For every free block, increment its “free” counter.
5. A correct block appears exactly once: either in use or free.

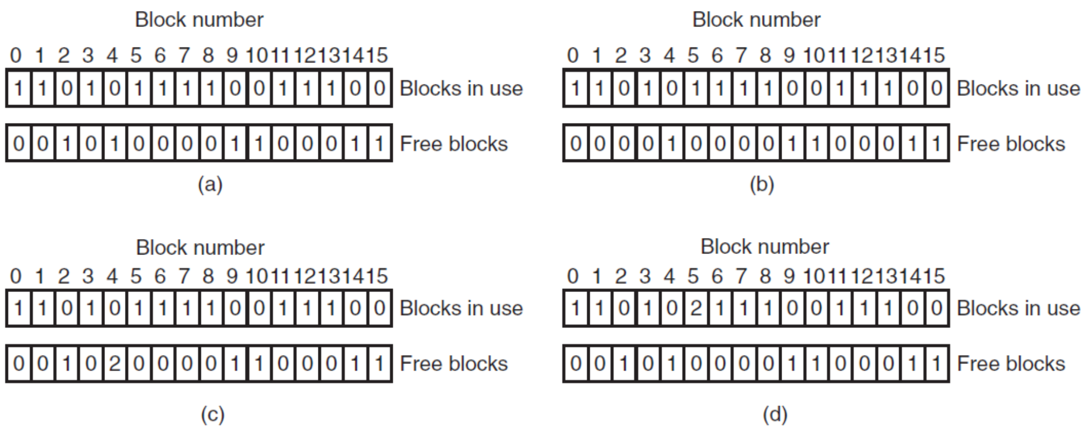

Figure 4-27. File-system states. (a) Consistent. (b) Missing block. (c) Duplicate block in free list. (d) Duplicate data block.

5 Block Error Cases

Case	Meaning	Fix
Consistent	Each block appears once, either used or free.	No repair.
Missing block	Block appears in neither table.	Add it to free list.
Duplicate free block	Block appears more than once in free list.	Rebuild free list.
Duplicate data block	Same block appears in two or more files.	Copy contents to a new block for one file, then report damage.

Worst Case

A duplicate data block is serious. If one file is deleted, the shared block may be freed even though another file still points to it.

6 Directory and Link-Count Check

Method

Count how many directory entries point to each i-node, then compare that count with the link count stored in the i-node.

- Start at the root directory.
- Recursively scan all directories.

- Increment the counter for each i-node found in a directory.
- Hard links count because they are real directory entries.
- Symbolic links do not increment the target file's count.

7 Link-Count Errors

Error	Consequence	Fix
i-node count too high	File may never be deleted, wasting disk space.	Set i-node link count to actual directory count.
i-node count too low	File may be freed while a directory still points to it.	Set i-node link count to actual directory count.

Danger

Too low is worse: deleting one name can reduce the i-node count to zero, freeing blocks that another directory entry still needs.

8 Other Suspicious Cases

Case	Why report it
Bad directory format	Directory entries should contain valid i-node numbers and names.
Invalid i-node number	Directory points outside the i-node table.
Strange permissions	Example: outsiders have more rights than the owner.
Huge directory	May indicate corruption or abnormal use.
SETUID root file in user directory	Possible security problem.

9 Protection from User Mistakes

Example

The command `rm *.o` removes object files, but `rm * .o` removes all files in the current directory and then complains that `.o` does not exist.

Recycle Bin Idea

Some systems move deleted files to a special directory first. Space is reclaimed only when files are actually deleted from that directory.

10 Final Summary

Summary

File-system checkers repair crash damage by verifying that every block is either used or free, and that directory references match i-node link counts.